Human Activity Recognition Using Geophone Sensor Data

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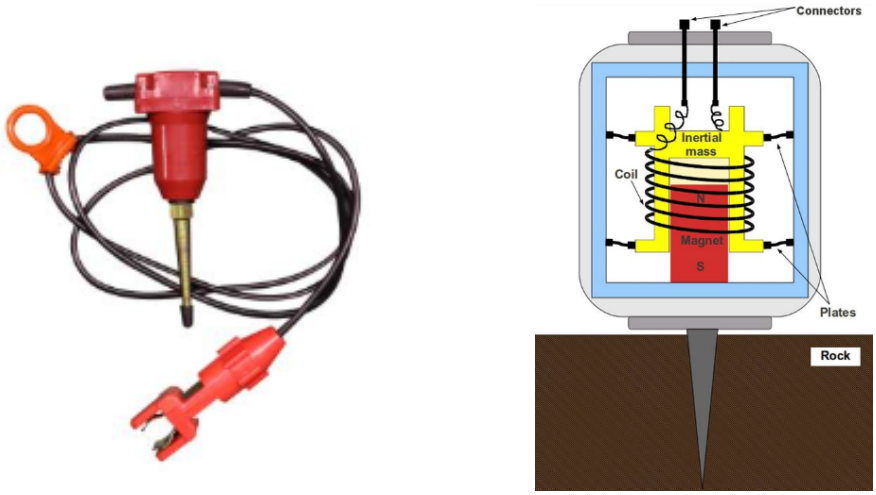

Figure 1: Images of a geophone (left) and its internal diagram (right) [5, 6].

Abstract

Human Activity Recognition (HAR) plays an increasingly important role in healthcare, safety, and intelligent environments. This report presents a data mining project that investigates whether geophone vibration signals can accurately classify the following three common human activities: standing, running, and walking. Using a Kaggle dataset [8] consisting of 1900 samples across fourteen statistical and frequency-domain features, we developed a complete data mining pipeline involving preprocessing, feature extraction, clustering evaluation, anomaly handling, and predictive modeling. A variety of algorithms and models were evaluated, such as Decision Trees and Logistic Regression, and compared against a Random Forest classifier. We additionally tested models using isolated statistical subsets of the data—such as only mean values, only standard deviations, or other individual features—to examine how specific attributes contributed to classification accuracy. We applied refinements and performed several rounds to evaluate results, and learned the significant impact of noisy participant data on model generalization and highlighted the limitations of frequency-based features in geophone recordings. Implemented in Python, the system demonstrates how statistical patterns in ground vibrations can be leveraged to distinguish human activities with strong accuracy. This work provides a comprehensive analysis of our methodology, technical contributions, and experimental findings, and discusses opportunities for future extensions such as anomaly detection and sleepwalking analysis.

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CCS Concepts

• **Computing methodologies → Supervised learning by classification**; • **Hardware → Sensor devices and platforms**; • **Information systems → Data mining**; • **Applied computing → Health informatics**.

Keywords

Human activity recognition; Machine learning; Geophone sensing; Random Forest; Signal Processing; Data Mining; Feature Extraction.

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1 Introduction

Human Activity Recognition (HAR) has emerged as a foundational component in modern intelligent systems, ranging from patient monitoring in healthcare settings to behavior recognition in smart homes and automated safety systems. Traditional HAR frameworks rely heavily on wearable sensors such as smartphone-based motion units. While these technologies can provide rich and accurate data, they also require direct physical contact with the user, constant battery consumption, and explicit user participation [7]. In contrast, geophones offer a passive, infrastructure-based alternative capable of detecting human activities without requiring any wearable devices. Geophones measure ground vibrations and are sensitive enough to detect subtle differences in movement patterns.

Our project seeks to explore whether geophone vibration data—represented not as raw signals but as aggregated time and frequency-domain features—can reliably distinguish between the three aforementioned activities: standing, walking, and running [1]. The problem is prevalent because vibration patterns of walking and running have overlapping characteristics–for example, a fast walk may produce mean vibrations nearly identical to those of a slow jog [2]. In addition, environmental noise, such as doors closing, HVAC systems, or other individuals walking in nearby areas can produce interference that distorts the signal. Further challenges may include differences in recording environments: indoor vs outdoor (surfaces). Altogether, these can further complicate classification. Early dataset exploration highlighted several challenges, including variable participant behavior, miscalibrated sensors, and low interpretability of certain frequency-derived features [2].

The data set used for this work consists of 1,900 samples, each representing a three-second window of geophone activity aggregated into 14 derived features [8]. These windows were collected from five individuals performing controlled repetitions of three movements. This design enables a structured multiclass classification task, while also allowing examination of participant-specific biases and inconsistencies. The inclusion of outlier participant data (identified as Alihan in Checkpoint 1) provided further opportunities for analyzing the robustness and generalizability of models [2].

1.1 Importance

The importance of this work extends beyond academic curiosity. If geophone-based HAR can achieve reliable accuracy using lightweight features and traditional machine learning methods [**?**], the approach could serve as a cost-effective and scalable alternative to wearable motion tracking systems. Potential applications include:

- **Monitoring elderly patients for falls or sleepwalking behaviors.** Geophone-based sensoring can offer a non-intrusive way to observe mobility patterns without requiring wearable devices, which is beneficial for the situations where elder adults may either refuse to or forget to wear them, or remove during sleep. Continuous vibration monitoring can detect sudden or abrupt changes that can correlate with falls, as well as identify unusual activity as wandering or sleepwalking, and overall enhance patient awareness and safety [2].
- **Enhancing smart home systems with context-aware automation.** Integrating activity recognition into smart homes enables more adaptive and personalized automation. For instance, detecting that a user is walking toward a specific room could trigger lighting adjustments, temperature changes, or appliance activation. Recognizing high-intensity movement, such as running, may allow the system to modify HVAC settings for comfort or safety. Because geophones function passively and require no user interaction, they support seamless integration into smart-home ecosystems while maintaining user convenience.
- **Identifying sudden transitions from walking to running as indicators of danger or emergency situations.** Abrupt changes in movement patterns often correspond to urgent or hazardous circumstances, such as fleeing from a threat, responding to an alarm, or experiencing distress. Geophone sensors can detect these rapid transitions in vibration intensity and frequency, enabling real-time alerts when a user who was previously walking suddenly begins running without context. Such detection capabilities can support emergency-response systems, safety monitoring, and real-time hazard recognition in public or private spaces [2, 10].
- **Detecting unauthorized movement in security-sensitive environments.** Facilities such as laboratories, server rooms, and restricted-access buildings require reliable detection of unexpected activity. Geophones can sense ground vibrations caused by footsteps, running, or forced entry attempts, even when individuals are not directly visible to cameras. Unlike video surveillance, vibration-based monitoring preserves privacy and can detect activity around corners, behind obstacles, or in low-light areas. This makes geophone-based HAR a practical component of multi-layered security systems.

1.2 Existing Literature, System Overview, and Data Collection

Our objective is to investigate several baseline classification algorithms, construct a robust preprocessing and modeling pipeline, and systematically evaluate model performance. In existing literature, most research on HAR relies on wearable sensors embedded in smartphones or smartwatches [11]. These studies have shown that high-dimensional sensor signals–combined with deep learning–can reliably distinguish between activities from walking, sitting, climbing stairs, and more. However, due to the fact that these devices require the users to wear them, it can limit practicality and adaptation to real-world deployments.

Our research takes a different perspective by exploring whether simpler, infrastructure-based geophone measurements–translated into statistical and frequency-domain features–can support accurate activity classification. We use a publicly available dataset of 1900 rows of samples, each representing a fixed 3-second window and summarized with fourteen features, including but not limited to: mean, standard deviation, amplitude, and dominant frequency [8]. These features feed into our pipeline, which consists of data cleaning/preprocessing, clustering-based anomaly detection, feature selection, baseline and ensemble modeling, and thorough evaluation.

In our experiments, we compare classical machine learning algorithms (Logistic Regression, Decision Trees, K-Nearest Neighbors, Gaussian Naïve Bayes, Linear SVM) to a tuned ensemble model (Random Forest). Our results reveal important insights: noisy data from a miscalibrated participant dramatically degrades performance, while refined preprocessing and feature-focused modeling significantly improves accuracy. We also observe that many of the frequency-domain features, which are often assumed beneficial in wearable-sensor HAR, offer limited predictive value in geophone-based vibration data. Ultimately, our pipeline shows that a well-tuned Random Forest classifier can outperform simpler methods under these constraints — supporting the feasibility of non-wearable HAR systems [2, 3].

2 Definitions and Problem Statement

2.1 Important Definitions

The dataset used in this study provides a structured transformation of raw geophone vibration measurements into interpretable numerical descriptors suitable for machine learning [8]. Below, we define the key concepts, variables, and components relevant to the classification task.

Data Instance: A single data instance corresponds to a fixed-length time window of approximately three seconds of geophone recordings. Each window consists of 120 raw vibration readings collected sequentially. These readings are then aggregated into a set of statistical and frequency-domain features that summarize the behavior of the vibration signal during that interval.

Prediction Target: The goal of the classification task is to predict the human activity occurring within the observed time window. Each data instance is labeled as one of the following classes:

$$y \in \{\text{standing}, \text{walking}, \text{running}\}$$

where standing corresponds to low-intensity, nearly motionless vibration signals; walking corresponds to moderate, periodic vibration patterns; and running corresponds to high-intensity, high-variation signals.

Feature Set: Each sample is represented by fourteen numerical features derived from the original 120-point vibration sequence [8]. These features fall into two main categories:

- *Time-domain statistical features:*
  - **Mean**: The average signal value over the 3-second window, capturing overall vibration intensity.
  - **Median**: The central value of the signal distribution, providing robustness against outliers.
  - **Standard Deviation**: A measure of variability indicating how widely the signal deviates from its mean. Higher values typically correspond to more dynamic activities like running.
  - **Minimum**: The smallest recorded vibration value within the window.
  - **Maximum**: The largest recorded vibration value within the window, often reflecting peak movements.
  - **Range**: Defined as max − min, representing the spread of the signal amplitude.
  - **First Quartile (Q1)**: The 25th percentile of the distribution; useful for describing lower-end vibration intensities.
  - **Third Quartile (Q3)**: The 75th percentile; useful for describing upper-end intensity.
  - **Skewness**: A measure of distribution asymmetry, showing data skewed to the left or right.
- *Frequency-domain features:*
  - **Dominant Frequency**: The frequency associated with the repetition of the movement. For example, faster footsteps often result in higher dominant frequencies.
  - **Signal Energy**: The total energy of the signal, computed as the sum of squared amplitudes. It reflects the overall intensity and is typically higher in running compared to walking or standing.

2.2 Dataset Overview

The dataset was collected from five individuals—Furkan, Enes, Yusuf, Alihan, and Emir—performing repeated instances of three core activities: walking, running, and standing still (referred to as "waiting" in the raw dataset). Each individual contributed multiple 3-second windows, resulting in approximately:

- **Total Samples:** 1900
- **Samples per Participant:** 360
- **Participants:** 5
- **Activities:** Walking, Running, Standing
- **Sampling:** 120 raw vibration points per 3-second window
- **Features:** 14 per sample (see previous section)

The data is recorded in a controlled environment but is still subject to individual variability and occasional noise. One participant's data (Alihan) differs noticeably from others, indicating possible sensor miscalibration [2]. This characteristic makes the dataset particularly valuable for studying robustness, anomaly detection, and the effects of noisy data on machine learning model performance.

2.3 Formal Problem Statement

The problem addressed in this project involves determining the type of human activity occurring within a short, three-second vibration window recorded by a geophone sensor. The data provided for each sample includes fourteen numerical features that summarize the statistical and frequency-based behavior of the vibration signal. The objective is to develop a data pipeline model that can use these features to accurately classify each window as standing, walking, or running.

This task comes with several constraints. First, some activities share similar vibration patterns, such as fast walking and slow running, which reduces the separability between classes. Second, the dataset contains noise and inconsistencies, including a participant whose measurements exhibit characteristics that are highly inconsistent with the rest of the dataset. Third, vibration measurements can be affected by environmental factors such as indoor versus outdoor surfaces, nearby movement, or other external noise sources. These challenges make the classification problem more complex and require a robust modeling pipeline to achieve reliable results.

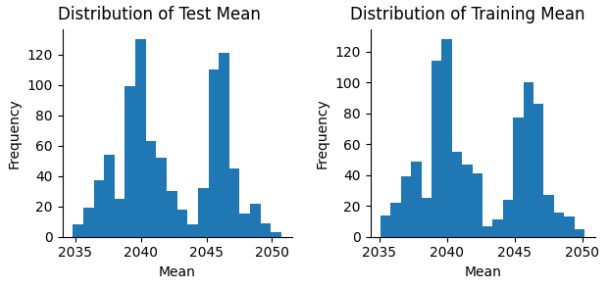
3 System Overview

Our pipeline, illustrated conceptually in the project slides [2], consists of five major components designed to transform raw sensor values into accurate predictions:

3.1 1. Data Sampling and Partitioning

We begin by creating training/testing partitions via stratified random sampling. As shown in Checkpoint 1 distributions [2], the mean-value histograms of both splits were compared to ensure that no partition disproportionately favored certain movement intensities or participant patterns. This step was essential to avoid model bias and to guarantee reproducible performance across random seeds. The data sampling process began with the generation of testing and training datasets. We then randomly split the dataset into two identically sized partitions. We compared the distribution

of the mean feature across the split to ensure that there was no disproportionate bias among the partitions.



3.2 2. Clustering Analysis

Even though labels were provided, clustering analysis served two purposes:

(1) Identify naturally occurring feature groupings.
(2) Detect anomalies or participant-level inconsistencies.

Clustering visualizations revealed that one participant, Alihan, produced measurements significantly outside the expected range, raising concerns regarding sensor miscalibration. We found that after checkpoint one, this clustering did not assist in the understanding and calculation of the dataset. We focused primarily on feature generation and linear regression. In [3], we determined that the best route forward was to specifically focus on assisting random forest-style predictors since it was our best-performing predictor. We ran RFE Feature Selection, Extra Trees Classifiers, AdaBoost + RF, and advanced stacking to assist with random forest

3.3 3. Feature Extraction and Selection

The dataset already included derived features, but part of our work involved evaluating their statistical distributions and correlations. We identified that the mean value of vibration signals consistently served as a strong indicator of activity intensity, whereas frequency-domain features demonstrated inconsistent patterns [2]. This prompted additional emphasis on time-domain features during model tuning. The derived data in the dataset is messy and ill-defined. The Kaggle dataset [8] provides no analysis as to how the data was generated. The Kaggle page does not provide units of measurement. Some information within the dataset requires no interpretation, such as the timestamp and user, but other features paint a blurry picture without context. The mean feature does not indicate units of measurement, leading us to have to deduce the context behind it. Since the other features are based off this nebulous data point, we must draw conclusions based off assumptions rather than defined evidence. To do this we generated features from the dataset and manipulated the weight of the data to assist in random forest-style predictors.

3.4 4. Predictive Modeling

A range of baseline models were selected based on their complementary properties:

- Linear models for interpretability.
- Tree-based models for nonlinearity handling.
- Probabilistic and distance-based models for comparative analysis.

Random Forest ultimately became the primary model after outperforming competitors in both accuracy and confusion matrix clarity. Through experimentation we determined that models based around random forest-like XGBoost may also be advantageous given our feature manipulation. The new models selected were

- XGBoost
- Tuned Random Forest
- Voting Ensemble
- Gradient Boosting
- Logistic Regression
- Random Forest + Linear Regression
- Random Forest + Linear Regression (Residual Refinement)
- RF with RFE Features Accuracy
- Extra Trees Test Accuracy
- AdaBoost + RF Test Accuracy

3.5 5. Evaluation

The evaluation stage involved computing accuracy, precision, recall, F1-score, and confusion matrices. Our pipeline also included analyzing feature importance shifts over time and re-evaluating models after excluding or adjusting outlier data. The evaluation process will be comparing the accuracy rates of each predictive process. We will be running approximately 200 estimators for each random forest-based estimator.

4 Technical Details

4.1 Feature Extraction

The dataset includes both time-domain and frequency-domain features. Time-domain statistics such as mean, standard deviation, and range capture amplitude characteristics of movement. Walking typically exhibits moderate mean values and variance, whereas running generates higher amplitude signals with greater variability. Unfortunately, we do not know what exactly is being expressed in the time-domain statistics. We can assume that the mean is found to be an expression of geophone data in millivolts, as geophone data is collected in voltage, but without a way to confirm it, we have to operate off of assumptions. Initially, we assumed that the data generated by user Alihan was fully incorrect and unable, but through the generation of the feature `mean_to_range_ratio` we were able to utilize the data.

Frequency-domain features, including dominant frequency and spectral energy, were initially hypothesized to differentiate between walking and running due to the periodicity of footsteps. However, exploratory analysis revealed:

- Dominant frequency values lacked consistency across samples.
- Signal energy correlated weakly with frequency.
- Frequency units and calibration were not documented, reducing interpretive reliability.

Despite these limitations, frequency features were retained to preserve dimensional variety, though time-domain features ultimately dominated model importance.

4.2 Predictive Modeling Approaches

4.2.1 Logistic Regression. Logistic regression was our general baseline. We did not anticipate logistic regression to fare particularly well, and at a top point trained accuracy of 34% we did not spend considerable time on this model. Logistic Regression existed to mostly define the baseline so we could develop out further.

4.2.2 Decision Tree. With logistic regression performing so poorly, we decided to develop out non linear approaches. The easiest of which to implement was the Decision Tree. This model was implemented after we learned about it in class. The accuracy of the the Decision Tree model doubled from Logistic Regression, indicating the direction we should be working towards.

4.2.3 K-Nearest Neighbors. KNN saw a marked decrease after feature engineering. During the first checkpoint we yielded relatively high results for KNN, however, this seemed to be an artifact due to Alihan's data. When we negated the outlier effect of Alihan's data, the K-Nearest Neighbor dropped to sub 50% levels.

4.2.4 Gaussian Naïve Bayes. Naïve Bayes performed poorly (44.11%), confirming that the independence assumption does not hold for correlated time-domain features.

4.2.5 Support Vector Machine. Linear SVM performed the worst out of the the entire initial experimental set. We see this as a reinforcement that linear decision making is the wrong format to pursue.

4.2.6 Random Forest. Random Forest consistently produced the highest accuracy, both before and after refinements. It handled noise reliably and produced results that interact with the features clearly. Random Forest had a gap of nearly 20% better than the rest of the early experimentation. We decided to further drill down on increasing the accuracy of Random Forest by employing enhancement techniques.

4.2.7 Random Forest Enhancement: Linear Regression (Stacking). A meta learning approach was to combine Random Forest and Linear Regression. Our initial hypothesis was that this would not provide a meaningful increase to our results as the dataset does not interact with linear modeling well.

4.2.8 Random Forest Enhancement: Random Forest + Linear Regression (Residual Refinement). An alternative format to match the performance of linear regression and random forest. This selection essentially exists as a sanity check to ensure that we are not making catastrophic errors with Random Forest.

4.2.9 Random Forest Enhancement: Extra Trees Classifier. Extremely Randomized Trees (Extra Trees) introduce additional randomness in both feature selection and split thresholds during tree construction. We implemented this to reduce overfitting and variance, even though it may cause some increased bias.

4.2.10 Random Forest Enhancement: AdaBoost with Random Forest. Adaptive Boosting sequentially trains multiple Random Forest base learners. AdaBoost teaches further iterations off of the misclassifications of past iterations. We implemented this predictor to ensure that misclassification errors are being handled well and that we can utilize the entirety of the dataset including previously seen misclassifications.

4.2.11 Random Forest Enhancement: Advanced Stacking (RF + Extra Trees + SVM). To hedge our bets even further, we implemented a stacking ensemble combining features from Random Forest, Extremely Random Trees, and Supper Vector Machines. Understanding the complexity of the dataset we indented to create a predictor that utilizes multiple strengths while reinforcing random forest.

4.2.12 Random Forest Enhancement: Recursive Feature Elimination. Recursive Feature Elimination (RFE) systematically removes less important features to create the most efficient dataset possible. We implemented this strategy with the assumption that we are not wrangling the data based off of its features in the most efficient manner.

4.2.13 XGBoost. XGBoost (eXtreme Gradient Boost) is a type of ensemble learning not dissimilar to RFE, however, XGBoost utilizes decision trees to preform a process similar to Extra Trees. XGBoost starts with an initial base learner, calculates the errors, and trains the trees based off of the errors. This is similar to that found in AdaBoost, but the process more than misclassifications errors in mind [4].

5 Experiments

5.1 Data Description

The dataset contains roughly balanced class distributions across the three activities, with slight variations across participants. We found that even with the undefined data the distribution is relatively easy to model. The data provides key temporal and ordinal data points which were categorized into standing, walking, and running samples. Our data prediction process was utilizing those labels to form a prediction model based off data that does not posses the label of movement.

5.2 Evaluation Metrics

The following metrics were used:

- **Accuracy**: primary metric for model comparison.
- **Precision, Recall, F1**: to handle class-specific imbalances.
- **Confusion Matrix**: to identify misclassification patterns.

5.3 Baseline Model Results

Baseline performances from Checkpoint 1 are listed in Table 2. Performance disparities between linear and nonlinear models indicate significant nonlinear structure in the data.

From the data found in checkpoint one, we dedicated time and effort to produce stronger results for random forest.

We also evaluated confusion matrices for best performing models.

Model	Accuracy (%)
Random Forest	75.19
KNN	73.70
Decision Tree	71.48
Gaussian NB	60.37
Logistic Regression	54.44
Linear SVM	47.78

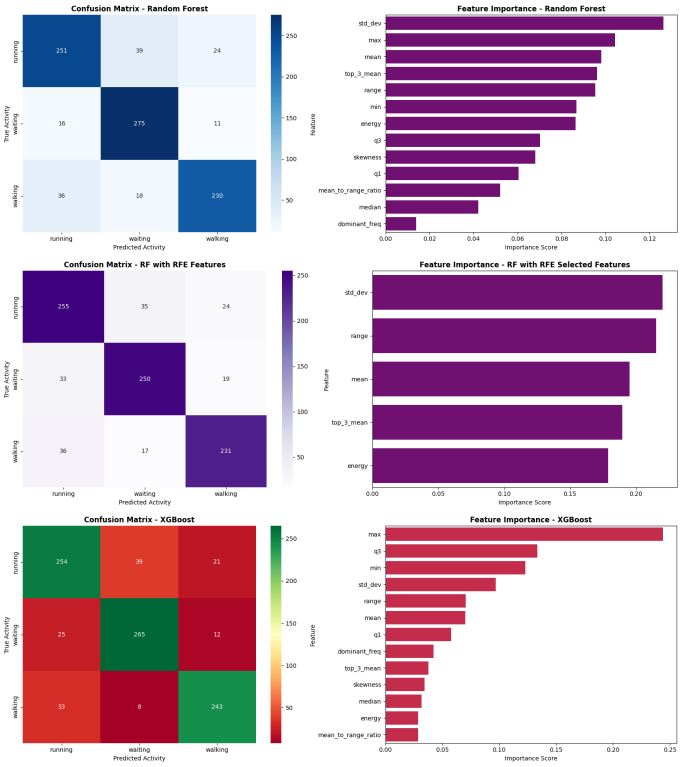
Table 1: Baseline Model Comparison (Checkpoint 1).

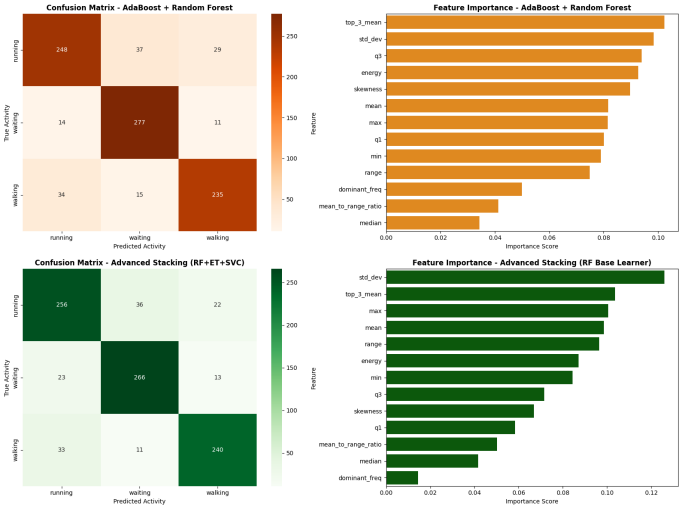
Model	Accuracy (%)
Advanced Stacking (RF+ET+SVC)	84.67
XGBoost	84.67
Random Forest (All Features)	84.00
Tuned Random Forest	84.33
Voting Ensemble	84.33
AdaBoost + RF	84.44
Baseline Tuned RF	84.33
RF + LR Stacking	84.33
Random Forest (Custom Features: 6)	83.67
Extra Trees Classifier	83.44
RF with RFE Features	81.78
Logistic Regression (Baseline)	75.19
RF + LR Residual	36.78

Table 2: Baseline Model Comparison (Checkpoint 2 and final comparison).

Table 3: Classification Performance Comparison Across Different Models

Model	Class	Precision	Recall	F1-Score	Support
Extra Trees	running	0.81	0.80	0.81	314
	waiting	0.83	0.88	0.85	302
	walking	0.86	0.83	0.85	284
	Acc.		0.83		900
AdaBoost + RF	running	0.84	0.79	0.81	314
	waiting	0.84	0.92	0.88	302
	walking	0.85	0.83	0.84	284
	Acc.		0.84		900
Stacking (RF+ET+SVC)	running	0.82	0.82	0.82	314
	waiting	0.85	0.88	0.87	302
	walking	0.87	0.85	0.86	284
	Acc.		0.85		900
RF with RFE	running	0.79	0.81	0.80	314
	waiting	0.83	0.83	0.83	302
	walking	0.84	0.81	0.83	284
	Acc.		0.82		900
XGBoost	running	0.81	0.81	0.81	314
	waiting	0.85	0.88	0.86	302
	walking	0.88	0.86	0.87	284
	Acc.		0.85		900
RF (Custom Features)	running	0.83	0.82	0.82	314
	waiting	0.83	0.87	0.85	302
	walking	0.85	0.82	0.83	284
	Acc.		0.84		900





We also observed f1 scores for each of the formats. This is displayed in Table 3

5.4 Improved Pipeline and Random Forest Refinements

Our pipeline was an iterative process that mirrored the methods we learned in class. Our process started with a rudimentary implementation all of the inital predictive models we learned in class with no consideration of how they would relate to eachother. Throughout the development process of the pipeline we found that our specific dataset works best with non linear prediction, and to that extent, we designed our pipeline around random forest and its derivatives. We identified that errors were not being handled, and we applied further development on those areas.

5.5 Case Studies

Analysis of confusion matrices shows that the majority of misclassifications occurred between walking and running. The similarity in vibration signatures for brisk walking and slow running supports observations made in the pitch slides [1]. Standing was rarely confused with dynamic activities due to its low signal variance and mean.

6 Related Work

Most existing HAR research leverages accelerometer and gyroscope sensors, often integrated into smartphones or wearables. Previous studies demonstrate that such sensors capture movement dynamics with high fidelity, but require physical contact and user compliance. Although these approaches laid an important foundation, their performance was often constrained by variability in human movement, background complexity, and camera positioning.

Geophone-based HAR is less explored but offers several advantages:

- Passive monitoring without intrusive sensors.
- Potential for full-room or building-level tracking.
- Low cost and infrastructure efficiency.

The introduction of deep learning has significantly transformed the field of Human Activity Recognition. Convolutional Neural Networks (CNNs) have enabled more robust extraction of spatial patterns from sensor data, while Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have become widely used for modeling the temporal structure of movement sequences. These architectures allow HAR systems to capture both fine-grained spatial cues and long-range motion dynamics, consistently outperforming traditional machine-learning approaches across numerous benchmark datasets [13–15]. Our work aligns with these emerging trends by examining activity recognition beyond wearable sensors. In particular, it contributes to the expanding research on non-wearable sensing by exploring how vibration-based features from geophones correspond to human motion patterns [16].

Our work aligns with emerging studies on non-wearable sensing, particularly those exploring floor-based, structural, or environmental sensors as alternatives to body-mounted devices. While most prior research focuses on gyroscopes, or smartphone-embedded sensors, only some examine vibration-based signals captured from the environment itself. By analyzing how statistical and frequency-derived features from geophone measurements relate to human motion, our project contributes additional insight into this area. It helps bridge the gap between wearable-centric HAR methods and approaches, demonstrating that meaningful activity information can be extracted even from low-cost, passive vibration data. This introduces geophone-based recognition as a promising and practical direction for future HAR systems, especially in settings where wearables are inconvenient, intrusive, or unreliable.

7 Conclusion

Through comprehensive analysis, this work demonstrates that geophone-derived features can effectively support a multiclass HAR system. Despite the presence of noise, a limited dataset size, and inconsistencies in frequency-based measurements, our refined data mining pipeline produced strong and consistent classification performance, particularly when using Random Forest models. These findings directly support our original goal of determining whether geophone sensor data, combined with classical machine learning methods, can accurately distinguish between the three activities of standing, walking, and running. The results show that even with lightweight statistical features, it is possible to achieve meaningful accuracy and robust class separation under realistic conditions.

Among all the information gathered from this project, several key findings were observed.

- Time-domain features (mean, range, max) are the most reliable and informative indicators of movement.
- Frequency-domain features lack consistency, likely due to sensor calibration differences and environmental interference.
- Outlier detection and removal dramatically improve model generalization.
- Random Forest consistently offers the strongest balance between performance, interpretability, and robustness compared to other baseline models.

These observations reinforce the feasibility of using geophone sensors for practical activity recognition applications and demonstrate that such modeling pipelines can achieve meaningful results.

While these results demonstrate the potential of our approach, several opportunities remain for further exploration.

Future work in this area may incorporate:

- Real-time streaming classification pipelines with continuous monitoring.
- Advanced feature engineering or deep learning approaches, such as to capture richer patterns from raw geophone signals.
- Anomaly detection for the identification of sleepwalking, falls, or any abrupt activity changes.

These directions would not only build on the strengths demonstrated in this study but also push geophone-based activity recognition closer to practical deployment in healthcare, safety monitoring, and smart-environment systems.

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Source Code

https://github.com/SShar195/CSE572GroupProject